

ALI AHAD

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ABOUT

I am a Software Engineer at Google focusing on Security for Emergent AI Platforms. I hold a PhD in Electrical and Computer Engineering with a focus on systems and software security, specializing in program analysis, reverse engineering, and forensic techniques. Core Focus: *Security for Emergent AI Platforms • Program Analysis & Reverse Engineering • Systems & Software Security • Decompilation & Binary Hardening*

EDUCATION

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| University of Maryland – College Park, MD | Defended April 2026 |
| • PhD in Electrical and Computer Engineering | |
| University of Virginia – Charlottesville, VA | August 2023 |
| • MS in Computer Science | GPA: 4.0/4.0 |
| Lahore University of Management Sciences – Lahore, Pakistan | June 2020 |
| • BS in Computer Science | GPA: 3.9/4.0 |

EXPERIENCE

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| Google – Sunnyvale, CA | August 2026 – Present |
| <i>Software Engineer</i> | |
| • Security for Emergent AI Platforms. | |
| University of Maryland – College Park, MD | January 2024 – June 2026 |
| <i>Research Assistant</i> | |
| • Implement and evaluate novel malware analysis and information flow tracking methods; advance reverse engineering technologies to detect hidden malicious code and information leaks in real-world systems. | |
| • Developed FreePart, a partitioning technique for secure data processing (ASPLOS '24), and DeARM, an LLM-driven framework for improving ARM64 decompilation that outperforms SOTA decompilers by 10.2%. | |
| NVIDIA – Santa Clara, CA | May 2024 – August 2024 |
| <i>System Software Intern, Data Center System Security</i> | |
| • Enhanced TPM security for data center systems by implementing and validating DMTF/TCG TPM-over-SPDM protocols, prototyped on Virtual TPM (vTPM) for secure measurements and communications. | |
| NVIDIA – Santa Clara, CA | September 2023 – January 2024 |
| <i>Software Intern, Security</i> | |
| • Engineered a TPM-based attestation solution for server platform security, implementing EK/AK provisioning, RIM generation, and evidence reporting using the RATS architecture. | |
| University of Virginia – Charlottesville, VA | August 2020 – August 2023 |
| <i>Research Assistant</i> | |
| • Published four papers in top-tier venues in Security/Systems (CCS '21, FSE '21, S&P '22, S&P '23). | |
| • Led a cross-institutional project resulting in a first-author S&P '23 publication; mentored two undergraduate interns (Amazon Summer '22 & Appian Summer '23), guiding research and project development. | |

PUBLICATIONS

- ["FreePart: Hardening Data Processing Software via Framework-based Partitioning and Isolation"](#): Ali Ahad, Gang Wang, Chung Hwan Kim, Suman Jana, Zhiqiang Lin, and Yonghwi Kwon, *In Proc. of the 29th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '24)*.
- ["PyFET: Forensically Equivalent Transformation for Python Binary Decompilation"](#): Ali Ahad, Chijung Jung, Ammar Askar, Doowon Kim, Taesoo Kim, and Yonghwi Kwon, *In Proc. of the 44th IEEE Symposium on Security and Privacy (S&P '23)*.
- ["SwarmFlawFinder: Discovering and Exploiting Logic Flaws of Swarm Algorithms"](#): Chijung Jung, Ali Ahad, Yuseok Jeon, and Yonghwi Kwon, *In Proc. of the 43rd IEEE Symposium on Security and Privacy (S&P '22)*.
- ["Forensic Analysis of Configuration-based Attacks"](#): Muhammad Adil Inam, Wajih Ul Hassan, Ali Ahad, Adam Bates, Rashid Tahir, Tianyin Xu, and Fareed Zaffar, *In Proc. of the 29th Network and Distributed System Security Symposium (NDSS '22)*.
- ["Swarmbug: Debugging Configuration Bugs in Swarm Robotics"](#): Chijung Jung, Ali Ahad, Jinho Jung, Sebastian Elbaum, and Yonghwi Kwon, *In Proc. of 29th ACM SIGSOFT International Symposium on the Foundations of Software Engineering (FSE '21)*.
- ["Spinner: Automated Dynamic Command Subsystem Perturbation"](#): Meng Wang, Chijung Jung, Ali Ahad, and Yonghwi Kwon, *In Proc. of 28th ACM Conference on Computer and Communications Security (CCS '21)*.

RELEVANT COURSES

- **Program Analysis**: Software Analysis, Program Analysis, Compilers
- **Security**: Mobile & IoT Security, Network Security & Privacy, Cyber Forensics
- **Systems**: Computer Architecture, Operating Systems, Digital CMOS VLSI Design
- **Machine Learning**: Intro. to Artificial Intelligence, Machine Learning, Information Theory
- **Networks**: Internet Infrastructure, Network-Centric Computing

TECHNOLOGY

- **Languages**: Python, C, C++, Rust, BASH, Golang, Javascript, Dart
- **Frameworks**: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA, LLVM, Pandas, React, Flask
- **Reverse Engineering**: Ghidra, IDA, Angr, Uncompyle6, Decompyle3
- **Software Testing**: American Fuzzy Lop (AFL), KLEE, LibFuzzer
- **Infrastructure**: Git, Linux, Docker, SLURM, Wireshark

SKILLS

- **Security & Systems**: Systems & Software Security, Reverse Engineering, Program & Information Flow Analysis, Cyber Forensics, Secure System Design, Threat Modeling, Malware Analysis, Vulnerability Assessment
- **AI/ML & Program Analysis**: LLM Fine-Tuning & Specialization, Binary Analysis, Static & Dynamic Analysis, Algorithm Design & Optimization, Control Flow Analysis, Model Evaluation & Benchmarking
- **Systems Engineering**: Large-Scale Tool Development, Automated Testing & Fuzzing, Software Debugging
- **Research & Communication**: Cross-Functional Collaboration, Technical Roadmapping, Scientific Writing